Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 23-03-2026

RELATIONAL INVENTORY CONTROL & STOCK TRACKING SYSTEM

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

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INTRODUCTION

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- Many small businesses still manage inventory manually, leading to errors and inefficiency.
- Tracking products, stock levels, and updates becomes difficult without automation.
- This project is important to **digitize and simplify inventory management**.
- It reduces manual work and improves **accuracy and speed**.
- The system is useful for shops, warehouses, and small enterprises.
- It provides a **centralized platform** to manage product data.
- The system includes features like **Add, View, Update, Delete, and Search products**.
- Overall, it is a **web-based solution for efficient inventory control**.

PROBLEM STATEMENT

- Many businesses still use **manual methods** like notebooks or spreadsheets to manage inventory.
- This leads to **errors in data entry and stock calculations**.
- Tracking product details and availability becomes **time-consuming and difficult**.
- There is no proper system to **update or delete product information efficiently**.
- Searching for specific products is **slow and inconvenient**.
- Lack of automation results in **poor data management and inefficiency**.
- There is no centralized system to **store and access product data easily**.
- Hence, a **simple, automated inventory management system** is required.

OBJECTIVE(S)

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- To design and develop a **web-based Inventory Management System** for efficient product management.
- To implement features like **Add, View, Update, Delete, and Search products** to improve accuracy and reduce manual work.
- To maintain a **centralized database for product information**.

PROPOSED SYSTEM

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- The system is a **web-based Inventory Management System** with secure admin login.
- It allows users to **add, view, update, and delete product details** easily.
- A **dashboard interface** is provided for smooth navigation between modules.
- The system includes a **search feature** to quickly find products by ID or name.
- Data is stored in a **centralized MongoDB database** for efficient management.
- It reduces manual work by **automating inventory operations**.
- The system ensures **accuracy, consistency, and faster data processing**.
- It provides a **simple, user-friendly, and scalable solution** compared to manual systems.

PROPOSED SYSTEM ARCHITECTURE

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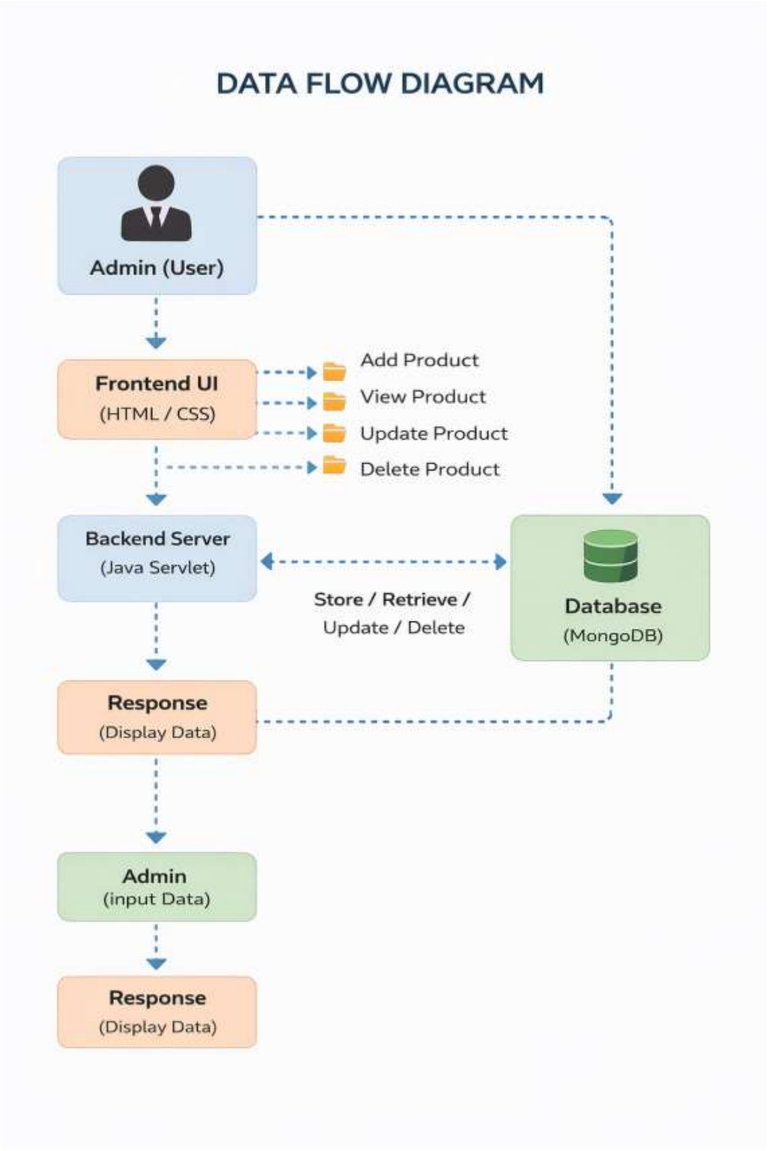
```
+------------------+
|  Admin (User)   |
+------------------+
        |
        v
+------------------+
|  Frontend       |
| (HTML, CSS)     |
+------------------+
        |
        v
+------------------+
|  Backend        |
| (Java Servlet)  |
+------------------+
        |
        v
+------------------+
|  Database       |
|  (MongoDB)      |
+------------------+
```

WORKFLOW


DATA FLOW DIAGRAM
Admin (User)
Frontend UI
(HTML / CSS)
Add Product
View Product
Update Product
Delete Product
Backend Server
(Java Servlet)
Store / Retrieve /
Update / Delete
Database
(MongoDB)
Response
(Display Data)
Admin
(input Data)
Response
(Display Data)

TECHNOLOGIES USED

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- **Frontend:**
  HTML , CSS

- **Backend:**
  Java(Servlet , JSP)

- **Database:**
  MongoDB

- **Tools:**
  Eclipse IDE , Apache Tomcat Server

- **Hardware Requirements:**
  Basic Computer/Laptop, Minimum 4GB RAM

- **Software Requirements:**
  JDK (Java Development Kit), Web Browser (Chrome/Edge), MongoDB Compass

MODULE DESCRIPTION

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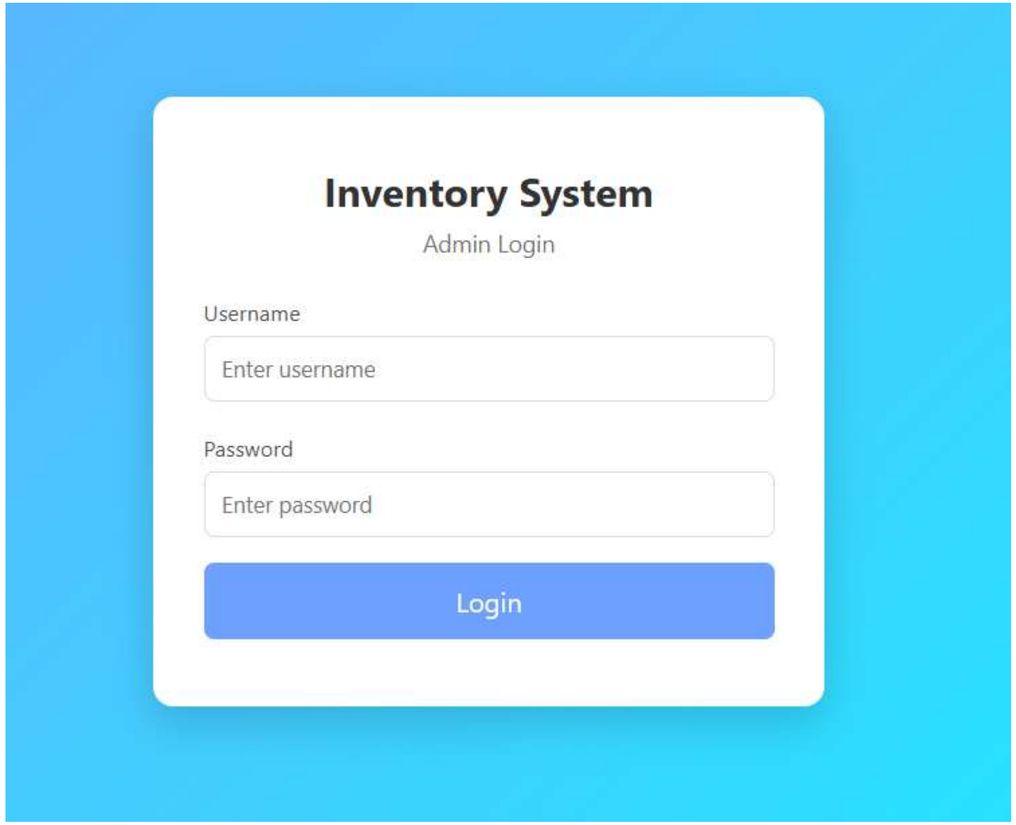
Module 1: Admin Login
Handles secure authentication of the admin using username and password.
Module 2: Dashboard
Provides a user-friendly interface with navigation options like Add Product, View Product, and Logout.
Module 3: Product Management
Allows admin to **add, view, update, and delete products** efficiently.
Module 4: Search Module
Enables searching products quickly using **product ID or name**.
Module 5: Database Management
Manages storing, retrieving, updating, and deleting product data in MongoDB.

IMPLEMENTATION SCREENSHOTS

Admin Page:

- Admin login with Username and Password.
- It open after entering system lock password .



IMPLEMENTATION SCREENSHOTS

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8(A)
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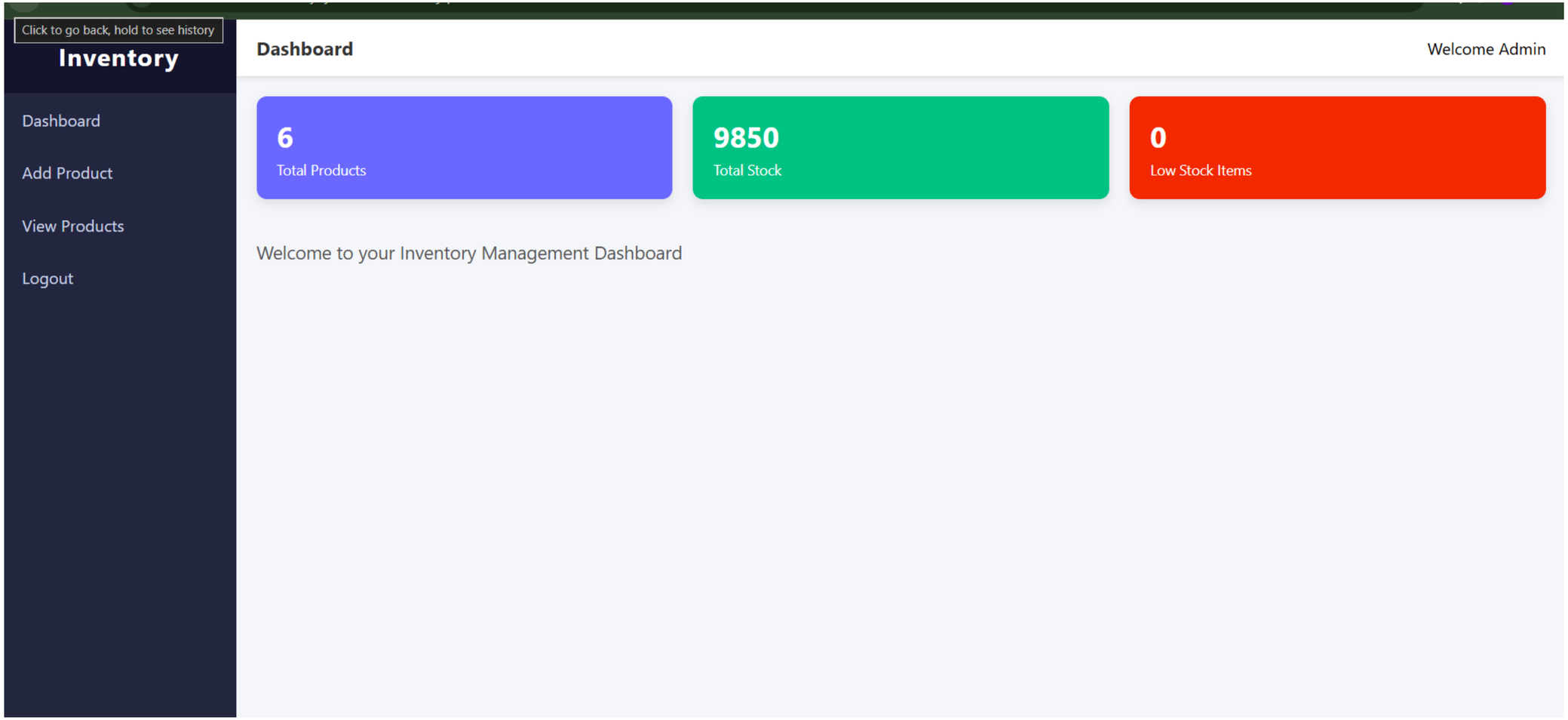
- **Home Page:**



IMPLEMENTATION SCREENSHOTS

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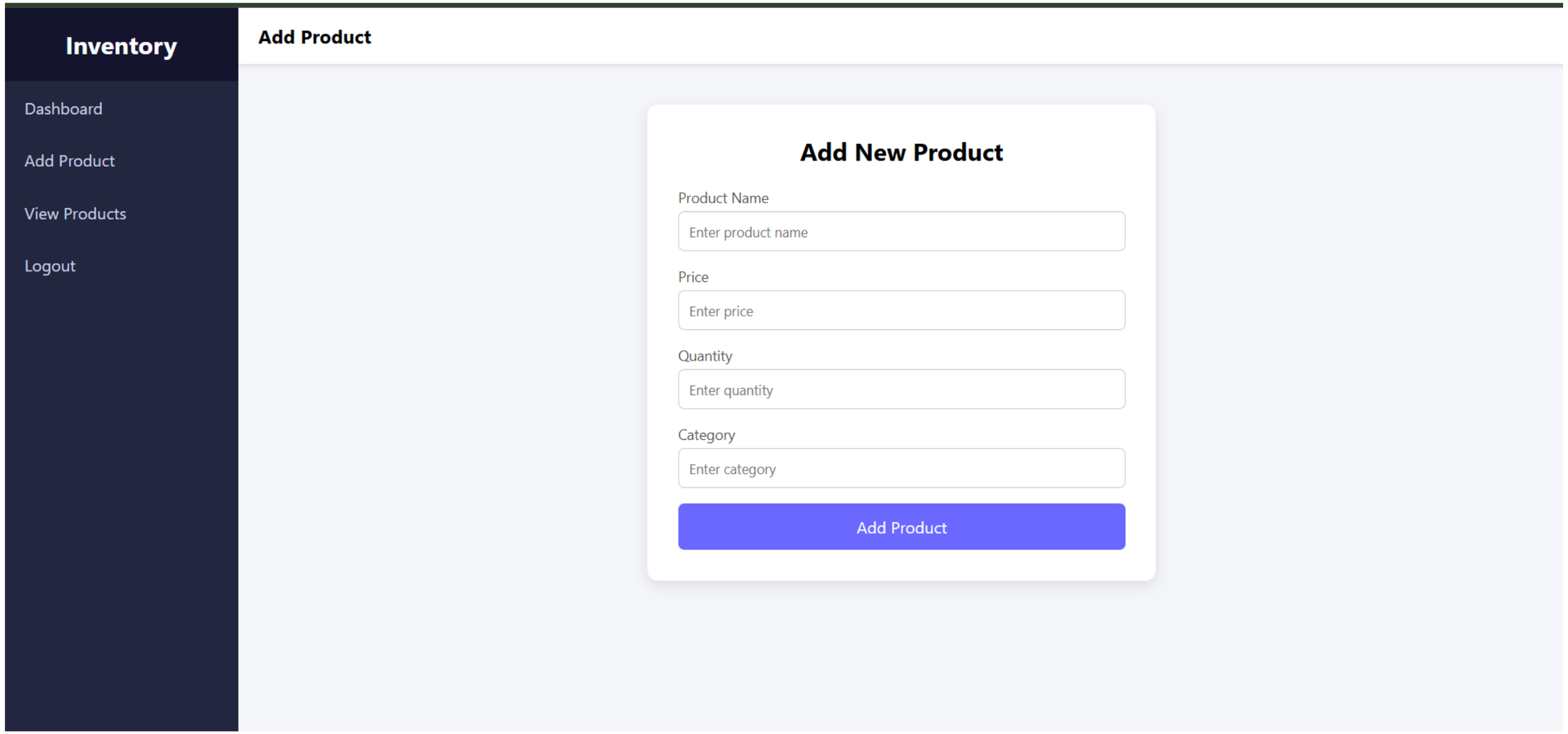
- **Add Product Page:**



IMPLEMENTATION SCREENSHOTS

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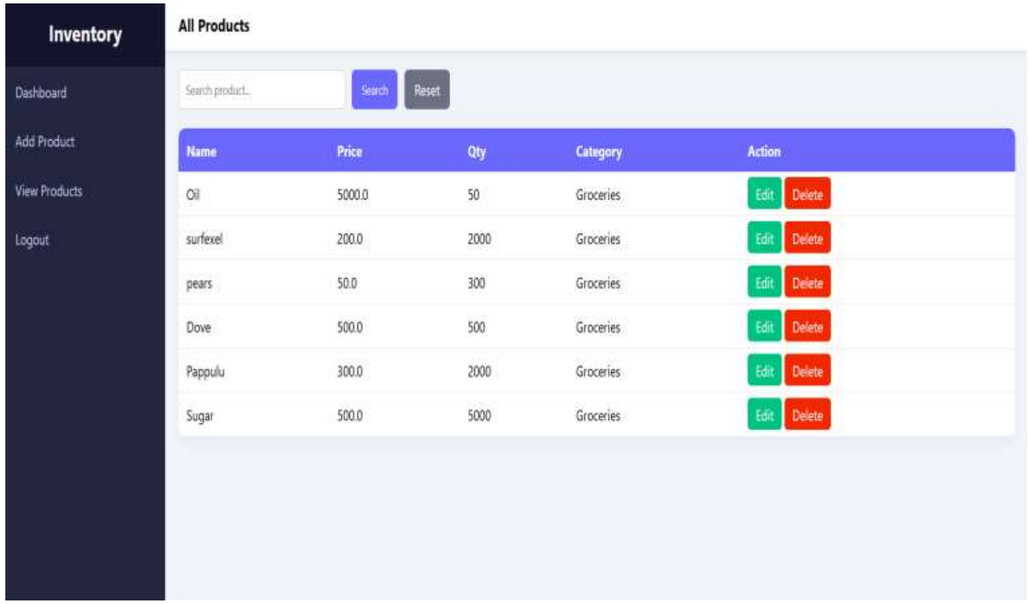
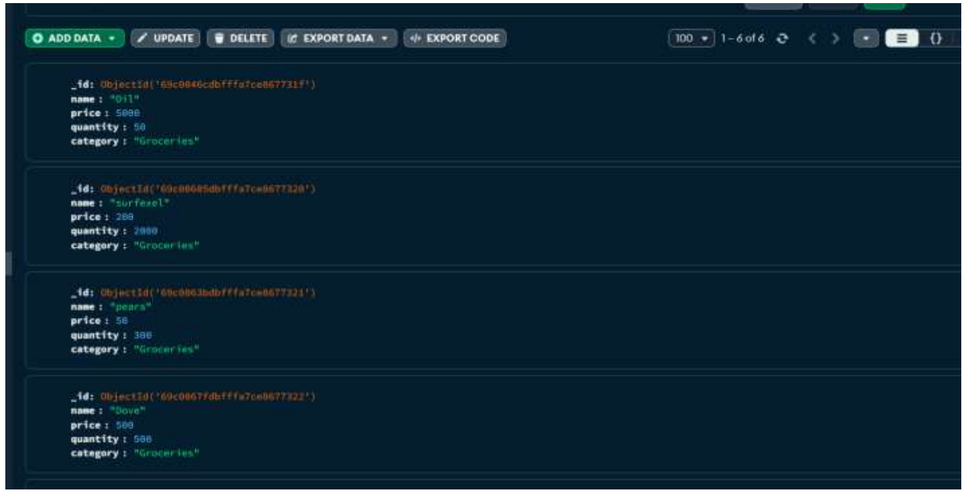
- **Product Page:**



-Database:



ADVANTAGES & LIMITATIONS

Advantages:

- Reduces **manual work and human errors**
- Provides **fast and efficient inventory management**
- Easy to **add, update, delete, and search products**
- User-friendly interface for easy operation
- Saves **time and effort**

Limitations:

- Only **admin access** (no multi-user roles)
- Basic UI (can be improved with better design)
- Requires **internet/local server setup**
- Limited features

Thank you